



Broadway Bound Consultations



RJ Clemens & Gary San Angelo



Mission Statement

“Our Mission is to help Broadway producers, investors, and artists maximize profitability, success, and to optimize investments with data driven insights. By leveraging historical trends, audience insights, and financial data, we will provide actionable strategies to maximize ticket sales, optimize show longevity, and improve overall theatre performance

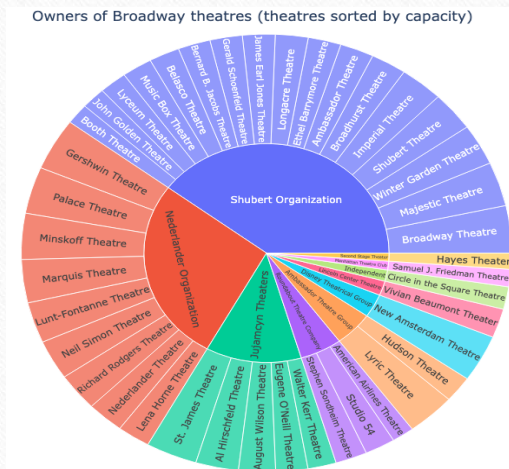


Services & Background Information

- Analyze ticket sales and audience engagement.
- Predictive modeling to help producers plan successful productions.
- Consulting on the marketing strategies based on past show performances.



Broadway Bound Consultations is a data analytics consulting firm for live theatre industry specializing in financial and performance analytics. Established in 2025, the company aims to provide insights to theatre professionals using historical Broadway data.



Data Gathering

- Broadway Bound Consultations sources historical performance data from industry archives, publicly available theater financial reports, and ticketing platforms.
- Data includes, ticket sales and pricing.
- Performance metrics such as seats sold, previews before shows and percent capacity.
- Show details such as show name and year.
- Capacity info such as seats in theater.
- The Broadway League
- Telecharge and Ticketmaster APIs
- Playbill
- Theatre venue reporting
- Marketing and attendance surveys



Data Cleaning & Processing

- ETL Procedure (Extract, Transform, Load)
- Raw .CSV files were loaded into Python for initial processing.
- All column names were reformatted for SQL conventions.
- Ensured data types were correct such as numerical and year fields.
- Missing values, such as show name entries, were replaced with “Unknown”.
- Duplicate rows were identified and removed.
- Cleaned datasets are imported to MySQL database, organized in relational tables following a EER schema structure.

Data Cleaning & Processing Continued

Calculations from data:

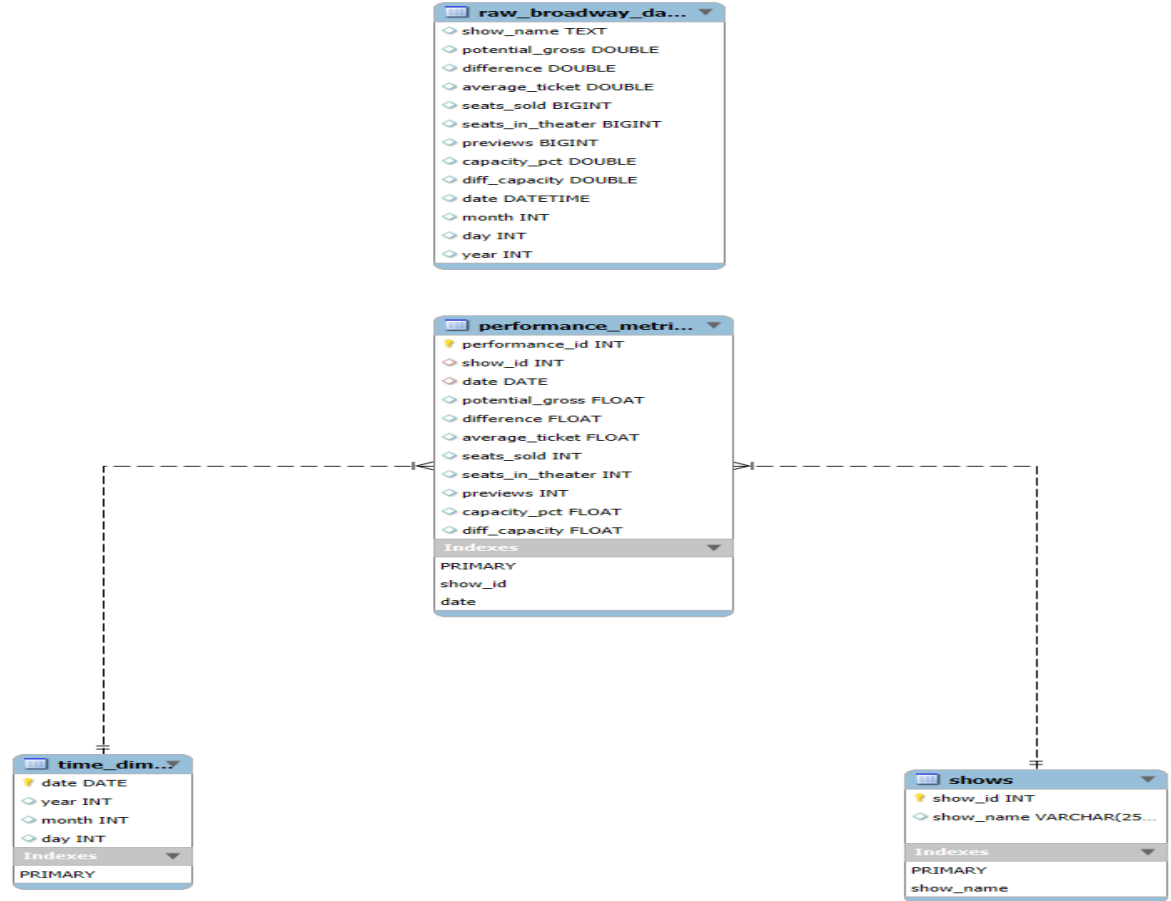
- $\text{Potential Gross} = \text{Seats Sold} * \text{Average Ticket Price}$
- $\text{Average Ticket} = \text{Potential Gross} / \text{Seats Sold}$
- $\text{Capacity} = (\text{Seats Sold} / \text{Seats in Theater}) * 100$
- $\text{Difference in Capacity} = \text{Capacity} - (\text{Previous Capacity})$

Database Storage & Usage

- Data is stored locally in a MySQL database.
- Weekly database backups are performed using mysqldump and cleaned datasets are stored on GitHub as external backups
- Database access is restricted to authorized employees within Broadway Bound Consultations, including data analysts and senior consultants.



Schema



Questions

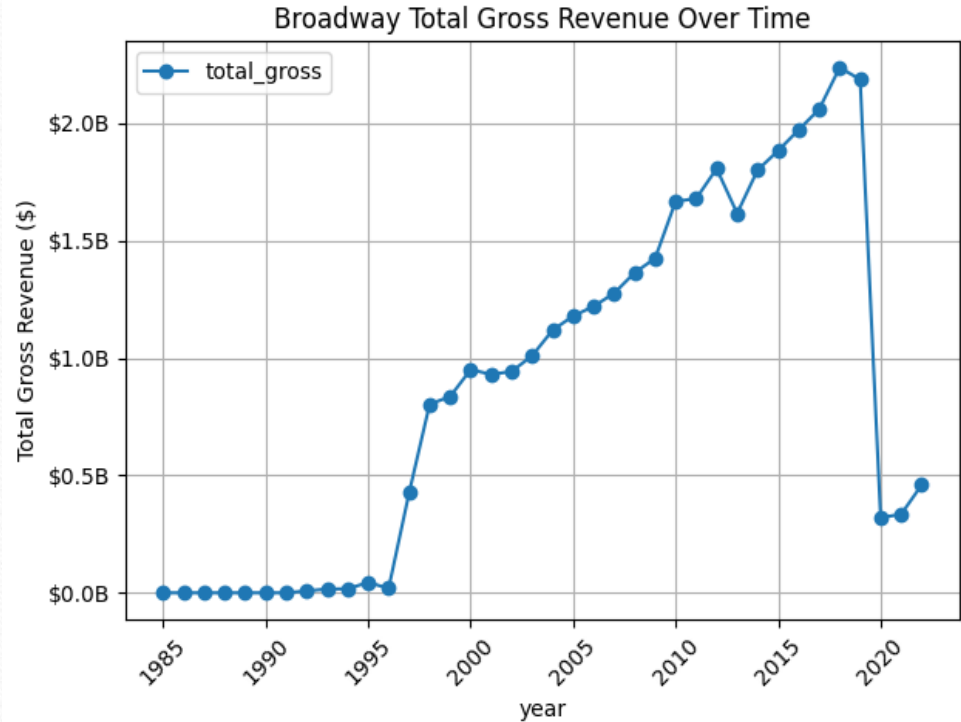
- How has Broadway's total gross revenue changed over time, and what does this indicate about the popularity of live theater?
- What shows have the highest seating capacity percentage?
- What are seasonal trends for Broadway shows?



Answers

```
SELECT t.year, ROUND(SUM(p.potential_gross), 2) AS total_gross
FROM performance_metrics p
JOIN time_dim t ON p.date = t.date
GROUP BY t.year
ORDER BY t.year ASC;
```

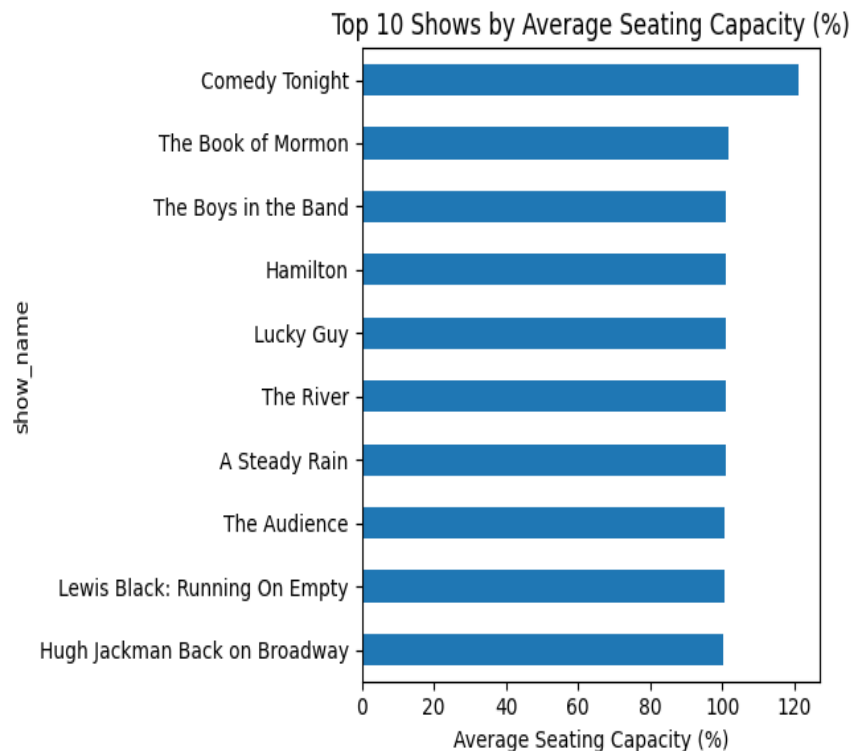
There is a consistent upward trend observed through the early 2010's indicating growing popularity and financial success of live theater. A sharp decline occurs around 2020 reflecting the COVID-19 pandemic's impact on live performances.



Answers

```
SELECT s.show_name, ROUND(AVG(p.capacity_pct), 2) AS avg_capacity
FROM performance_metrics p
JOIN shows s ON p.show_id = s.show_id
GROUP BY s.show_name
ORDER BY avg_capacity DESC
LIMIT 10;
```

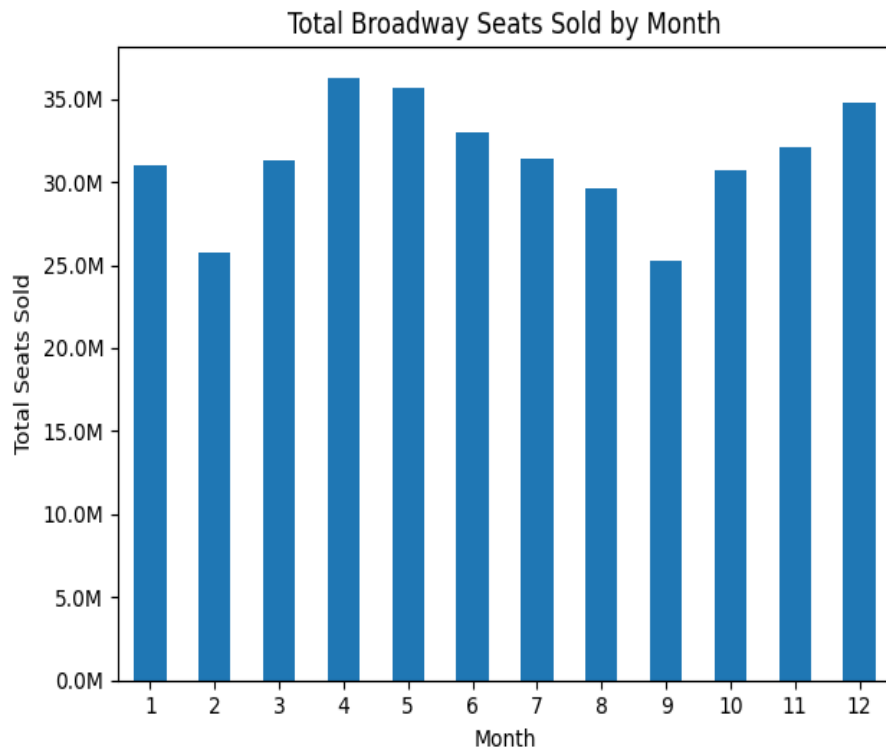
Top 10 Broadway shows ranked by their average seating capacity percentage. Most of these shows maintained near or above 100% capacity with 1 show exceeding capacity. This is possible due to standing room ticket sales or additional seating. These high percentages indicate a strong demand and consistent sell out performances.



Answers

```
SELECT t.month, ROUND(SUM(p.seats_sold), 0) AS total_seats_sold
FROM performance_metrics p
JOIN time_dim t ON p.date = t.date
GROUP BY t.month
ORDER BY t.month ASC;
```

Sales seem to peak during the spring (April-May) and holidays (November-December). There are noticeable declines during the summer months and January.



QR codes

Gary San Angelo



RJ Clemens

